# DOC334 Week 02

# Learning Outcome of This Session

- Data types
  - Numbers
  - Float values
  - Strings
  - Lists
  - Tuples
- Escape Characters
- Control structures
  - Sequence
  - Selections
  - Iterations

# Python Data Types

- Most other languages pre-set data type
  - int num = 5;
  - Dim num as Integer
- Python sets the type when you assign value
  - This can be dynamically changed
- X = 5
  - This becomes integer (int)
- The **type** will let you know the data type
  - type(X) will be int

# Python Data Types

- Text Type
  - str
- Numeric Types
  - int, float, complex
- Sequence Types
  - list, tuple, range
- Mapping Type
  - dict
- Set Types
  - set, frozenset
- Boolean Type
  - bool
- Binary Types
  - bytes, bytearray, memoryview

# Python Data Types (Our Scope in Classes)

- Text Type
  - str
- Numeric Types
  - int, float
- Sequence Types
  - list, tuple, range
- Boolean Type
  - bool

You are free to explore and use other types in ICW

# Samples

## Usage

- Value = ‘IIT’
- Value = “IIT”
- Value = 10
- Value = 55.63
- Value = [“hello”,”world”]
- Value = (“DOC333”,”DOC334”)
- Value = range(3)
- Value = False

## Check Type (in IDLE)

- Type(Value)
  - What do you see?
- Try this and figure the data type for each line of the left panel.
- Paste it to the Zoom chat

# Explicitly Set the Data Type

## Usage

- Value = str(10)
- Value = int(55.63)
- Value = float(20)
- Value = bool(5)
- Value = list((“A", “B", “C"))
- Value = tuple((“P", “Q", “R"))
- Value = int(False)

## Check Type (in IDLE)

- str(10)
  - What do you see?
- type(str(10))
  - What do you see?
- Try this and figure the data type for each line of the left panel.
- Paste it to the Zoom chat
- We’ll see what these are shortly!

# Casting

- Casting is the process of converting or specifying the data type you require
- This can be explicitly done
  - You did it in a previous slide!
- Python is an OOP language?
  - So, the answer may appear the word “class” in it
  - What is OOP? (extra reading for you)
- Needs some built-in functions to do the conversion
  - str(), int(), etc.

# Be Careful!

- X = 256.35
- Y = int(X)
  - What is the issue?
- What could go wrong?
  - Financial calculations
  - Scientific calculations, etc.



# Strings in Python

- Strings in Python is a series of characters
  - They may be alpha-numeric
  - Also, special characters
  - This can even be a one character!
- You can use “” or ‘’ for strings
- Triple quotes can be used for multiple string lines
  - “”” or ‘’’
- They are treated as arrays

# Sample String Values

- Try below in IDLE and look at the output
  - company_name = IIT
  - company_name = "IIT"
  - company_name = ‘IIT’
  - company_name = """Informatics

    Institute

    of

    Technology"""“
- What is “\n”?
  - Escape sequence
- name = "Nishan\nSaliya“
  - Check the value in IDLE and under a print() statement
  - What do you see?

# What is an Escape Character?

- These are special characters (or illegal charters) which you can not place directly in a string
- They are followed by a backslash “\”
- Most languages use these to display some alternative output
- Try the shown characters in the right side inside IDLE

- \’ for single quote
- \\ for backslash
- \n for new line
- \r for carriage return
  - What is this?
- \t for tab
- \b for backspace
  - Can you find other characters?

# Did All of Them work?

- Some will work in Console
  - Future lessons
- Try to write a code in IDLE to display the below output
- Once done, copy & paste your answer to Zoom chat as a direct message to me
  - DO NOT send to everyone chat

****** Dear Student ******
\\\\\\ Welcome to IIT \\\\\\

A tab space appear here

# Lists in Python

- One of the collections in Python
- They are,
  - Ordered
  - Changeable
  - Allow duplicates
  - Indexed
  - Allow multiple data items inside

# Try Below Exercise

- Try these in IDLE
  - sample_list = [1,200,305.6,'Hello',"IIT"]
  - sample_list[1]
- What should I type to get the output as “IIT”?
- What should I type to get the output as 305?
- What should I type to get the output as “H”?
- Send your output via Zoom chat

# Tuples in Python

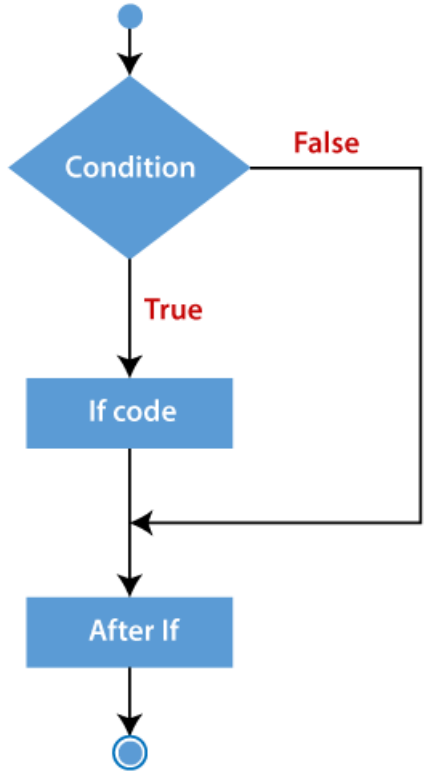
- One of the collections in Python
- They are,
  - Ordered
  - NOT Changeable
  - Allow duplicates
  - Indexed
  - Allow multiple data items inside

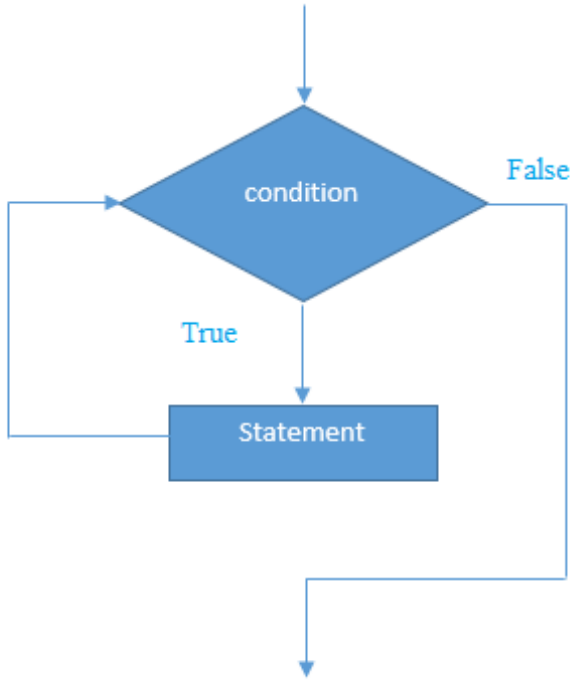
# Try Below Exercise

- Try these in IDLE
  - sample_tuple = (1,200,305.6,'Hello',"IIT")
  - sample_tuple[1]
- What should I type to get the output as “IIT”?
- What should I type to get the output as 305?
- What is the output for `sample_tuple[4] = 5`
  - Why?
- Send your output via Zoom chat

# Control Structures

- Python comes with 3 main control structures
  - Sequence
  - Selection
  - Iterations



# Sequence in Python

```
age = 18
print("You are”,age,”Years old”)
```

# Selection in Python

## NULL False Selection

```
age = 18
if age>=18:
    print("You are an adult")
```

## Complete IF

```
age = 10
if age>=18:
    print("You are an adult")
else:
    print("You are a child")
```

# Iterations in Python

- Python supports 2 loops structures
  - While loop
  - For loop

# While Loop in Python

```
count = 1
while count < 6:
  print(count)
  count += 1
```

- Python requires a variable to control the iterations
- The loop will run forever without the control variable

# For Loop in Python

```
for x in range(1,5):
  print(x)
```

```
R = range(1,5)
for x in R:
  print(x)
```

- These loops are used to iterate over a known sequence
  - List, tuple, string, etc.
- Due to this, control variable is not required

# For Loop in Python – With a List

```
my_list = ["hello","how","are","you?"]
for x in my_list:
  print(x)
```

No Control Variable

# For Loop in Python – With a Tuple

```
my_tuple = ("hello","how","are","you?")
for x in my_tuple:
  print(x)
```

No Control Variable

# For Loop in Python – With a String

```
my_string = "Informatics"
for x in my_string:
  print(x)
```

No Control Variable

# Question

1. Write a program to read your name and display it character by character.
2. Write a program to read a value from the user and tell them if it is a number or not
   - Characters are not allowed
   - Alpha-numeric are not allowed
   - Only numbers are allowed
3. Write a program to find the next odd number when any number is inserted. (Sample Marking Scheme)
   - If 5 is typed, answer is 7
   - If 6 is typed, answer is 7

# Essential Reading for you

- https://www.w3schools.com/python/python_datatypes.asp
- https://www.w3schools.com/python/python_strings.asp
- https://www.w3schools.com/python/python_lists.asp
- https://www.w3schools.com/python/python_tuples.asp
- https://www.w3schools.com/python/python_conditions.asp
- https://www.w3schools.com/python/python_while_loops.asp
- https://www.w3schools.com/python/python_for_loops.asp
  - Try the given exercise under each section

# Any Questions?